Estimate calculations by rounding

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: _________________

1. Read the model: Estimate 398,742 + 203,118 by rounding each number to the nearest 100,000. Copy or complete the first step.

2. Estimate 8,102 x 49 using convenient rounded numbers.

3. Estimate 623,590 - 188,420 to the nearest 100,000.

4. Miro says: Miro believes an estimate must equal the exact answer. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Estimate 398,742 + 203,118 by rounding each number to the nearest 100,000. Copy or complete the first step.
   Answer 400,000 + 200,000 = 600,000.

2. Estimate 8,102 x 49 using convenient rounded numbers.
   Answer 8,000 x 50 = 400,000.

3. Estimate 623,590 - 188,420 to the nearest 100,000.
   Answer 600,000 - 200,000 = 400,000.

4. Miro says: Miro believes an estimate must equal the exact answer. Is this correct? Explain.
   Answer An estimate is deliberately approximate and should show the likely size of the exact answer.

Estimate calculations by rounding

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ________________

1. Estimate 8,102 x 49 using convenient rounded numbers.

2. Estimate 623,590 - 188,420 to the nearest 100,000.

3. Choose an efficient estimate for 48,912 x 198 and explain the accuracy.

4. Identify and correct this misconception: Miro believes an estimate must equal the exact answer.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Estimate 8,102 x 49 using convenient rounded numbers.
   Answer 8,000 x 50 = 400,000.

2. Estimate 623,590 - 188,420 to the nearest 100,000.
   Answer 600,000 - 200,000 = 400,000.

3. Choose an efficient estimate for 48,912 x 198 and explain the accuracy.
   Answer About 50,000 x 200 = 10,000,000. Other justified close estimates are acceptable.

4. Identify and correct this misconception: Miro believes an estimate must equal the exact answer.
   Answer An estimate is deliberately approximate and should show the likely size of the exact answer.

5. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Estimate calculations by rounding

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Choose an efficient estimate for 48,912 x 198 and explain the accuracy.

2. Explain why this reasoning fails: Miro believes an estimate must equal the exact answer.

3. Create a new example that tests the same idea as: Estimate 623,590 - 188,420 to the nearest 100,000.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Choose an efficient estimate for 48,912 x 198 and explain the accuracy.
   Answer About 50,000 x 200 = 10,000,000. Other justified close estimates are acceptable.

2. Explain why this reasoning fails: Miro believes an estimate must equal the exact answer.
   Answer An estimate is deliberately approximate and should show the likely size of the exact answer.

3. Create a new example that tests the same idea as: Estimate 623,590 - 188,420 to the nearest 100,000.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.